Quadratics: Expanding & Factorising

Practice test 3 — answer sheet · 40 marks

Section A — Expanding double brackets

A1. $x^2 + 11x + 2x + 22$ (1) = **$x^2 + 13x + 22$** (1)

A2. **$x^2 - 49$** (1) the middle terms +7x and –7x cancel out (1)

A3. $4x^2 + 8x - 3x - 6$ (1) = **$4x^2 + 5x - 6$** (1)

A4. $x^2 + 6x + 9 + x^2 - 2x + 1$ (1) = **$2x^2 + 4x + 10$** (1)

Section B — Expanding three brackets

B1. $(2x - 1)(x + 3) = 2x^2 + 5x - 3$ (1) $(2x^2 + 5x - 3)(x - 2) = 2x^3 - 4x^2 + 5x^2 - 10x - 3x + 6$ (1) = **$2x^3 + x^2 - 13x + 6$** (1)

B2. $(x - 1)^2 = x^2 - 2x + 1$ (1) $(x^2 - 2x + 1)(x - 1) = x^3 - x^2 - 2x^2 + 2x + x - 1$ (1) = **$x^3 - 3x^2 + 3x - 1$** (1)

Section C — Factorising $x^2 + bx + c$ and the difference of two squares

C1. Multiply to 56, add to 15: 7 and 8 (1) **$(x + 7)(x + 8)$** (1)

C2. Multiply to –30, add to 1: 6 and –5 (1) **$(x + 6)(x - 5)$** (1)

C3. **$(10 + x)(10 - x)$** (1)

C4. $5(x^2 - 4x - 12)$ (1) = **$5(x - 6)(x + 2)$** (1)

Section D — Factorising $ax^2 + bx + c$

D1. $3 \times 10 = 30$; numbers: 6 and 5, so $3x^2 + 6x + 5x + 10 = 3x(x + 2) + 5(x + 2)$ (1) **$(3x + 5)(x + 2)$** (1)

D2. $4 \times -15 = -60$; numbers: –10 and 6, so $4x^2 - 10x + 6x - 15 = 2x(2x - 5) + 3(2x - 5)$ (1) **$(2x + 3)(2x - 5)$** (1)

D3. $(5x)^2 - (3y)^2$ (1) = **$(5x + 3y)(5x - 3y)$** (1)

Section E — Using factorising

E1. $x^2 - 6x + 8 = 0$ (1) $(x - 2)(x - 4) = 0$ (1) **$x = 2$ or $x = 4$** (1)

E2. Numerator = $3x(x + 2)$ (1) denominator = $(x - 3)(x + 2)$ (1) cancel $(x + 2)$: **$3x / (x - 3)$** (1)

Section F — Problem solving & proof

F1. $x^2 + 14x + 49 = (x + 7)^2$, so each side is $(x + 7)$ cm (1) perimeter = $4(x + 7)$ = **$(4x + 28)$ cm** (1)

F2. $(x + 2)(x + 5) = x^2 + 7x + 10$ (1) $x(x^2 + 7x + 10)$ = **$(x^3 + 7x^2 + 10x)$ cm^3** (1)

F3. $n^2 + (n + 1)^2 = n^2 + n^2 + 2n + 1 = 2n^2 + 2n + 1$ (1) $= 2(n^2 + n) + 1$, which is one more than an even number, so **always odd** (1)

F4. $(7.5 + 2.5)(7.5 - 2.5) = 10 \times 5 =$ **50** (1)

Marking note: the marks in brackets show where each mark is earned. Award method marks for a correct method even if there is a later slip. Accept answers in any equivalent form, e.g. terms in a different order, or brackets written the other way round: $(x – 7)(x + 4)$ is the same as $(x + 4)(x – 7)$. “Factorise fully” questions lose the last mark if a common factor is left inside a bracket.